Md Yousuf Harun

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Summary

Ph.D. candidate specializing in **continual or lifelong machine learning** of large-scale vision, language, and multimodal models. Published **7+ first-author papers** in top-tier venues including NeurIPS, ICML, CVPR-W, CoLLAs, and TMLR, addressing a core challenge in AI—**building models that remain adaptable, scalable, and trustworthy in dynamic environments**. Passionate about advancing **efficient, robust, and generalizable AI systems** for real-world deployment.

Education

Rochester Institute of Technology *Rochester, NY*
Ph.D. in Imaging Science *Aug. 2020 – Present*

- **Thesis**: Towards Efficient Continual Deep Learning
- **Advisor**: Dr. Christopher Kanan
- **Highlights**: 8+ publications on robust and efficient deep learning systems

University of Hawaii at Manoa *Honolulu, HI*
M.S. in Electrical Engineering *Aug. 2018 – May 2020*

- **Thesis**: Medical Image Segmentation using Deep Learning
- **Advisors**: Dr. Aaron Ohta and Dr. Il Yong Chun
- **Highlights**: 4 publications on deep learning for medical image segmentation

Khulna University of Engineering & Technology *Khulna, Bangladesh*
B.S. in Electrical & Electronic Engineering *Feb. 2012 – May 2016*

Experience

Armada Systems Inc *Rochester, NY (Remote)*
AI Intern *Jan 2026 – Apr 2026*

- Developed multimodal AI pipelines for video understanding using large vision–language models (VLMs), systematically evaluating and improving foundation models (e.g., Qwen3-VL) for safety and surveillance applications
- Built scalable training and evaluation frameworks for large multimodal models, leveraging parameter-efficient fine-tuning and domain-specific VQA dataset construction to enhance visual grounding, robustness, and performance on industrial video reasoning tasks

Rochester Institute of Technology *Rochester, NY*
Graduate Research Assistant *May 2021 – Present*

- Developed efficient continual deep learning algorithms for real-world applications
- Addressed core challenges in AI such as catastrophic forgetting, computational efficiency, and forward transfer
- Worked on open-world learning, neural collapse, OOD generalization, LLMs, MLLMs, VLMs, and foundation models

University of Hawaii at Manoa *Honolulu, HI*
Graduate Research Assistant *Aug. 2019 – May 2020*

- Conducted research on deep learning–based medical image segmentation and MRI reconstruction
- Designed AI-based image segmentation software for detecting genetically abnormal cells, enhancing diagnostic accuracy and supporting clinical workflows at the Pacific IVF Institute

Best Poster Award

- **M.Y. Harun**, J. Gallardo, and C. Kanan. Prioritized Training on Rehearsal Samples for Efficient Online Continual Learning. *In: IEEE Western NY Image and Signal Processing Workshop (**WNYISPW**)*, 2023 Link

Peer-Reviewed Publications

[$*$ denotes equal contribution]

1. J. Chen, **M.Y. Harun**, and C. Kanan. Unlocking ImageNet's Multi-Object Nature: Automated Large-Scale Multilabel Annotation. *In: Conference on Computer Vision and Pattern Recognition (**CVPR**) Findings*, 2026 Link
2. **M.Y. Harun**, J. Gallardo, and C. Kanan. Controlling Neural Collapse Enhances Out-of-Distribution Detection and Transfer Learning. *In: International Conference on Machine Learning (**ICML**)*, 2025 [26.9% accept rate] (Oral at ICCV 2025 T2FM Workshop) Link
3. **M.Y. Harun** and C. Kanan. A Good Start Matters: Enhancing Continual Learning with Data-Driven Weight Initialization. *In: Conference on Lifelong Learning Agents (**CoLLAs**)*, 2025 Link
4. S. Srivastava, **M.Y. Harun**, R. Shrestha, and C. Kanan. Improving Multimodal Large Language Models Using Continual Learning. *In: Conference on Lifelong Learning Agents (**CoLLAs**)*, 2025 Link
5. **M.Y. Harun***, K. Lee*, J. Gallardo, G. Krishnan, and C. Kanan. What Variables Affect Out-of-Distribution Generalization in Pretrained Models? *In: Neural Information Processing Systems (**NeurIPS**)*, 2024 [25.8% accept rate] Link
6. **M.Y. Harun**, J. Gallardo, J. Chen, and C. Kanan. GRASP: A Rehearsal Policy for Efficient Online Continual Learning. *In: Conference on Lifelong Learning Agents (**CoLLAs**)*, 2024 Link
7. **M.Y. Harun** and C. Kanan. Overcoming the Stability Gap in Continual Learning. *In: Transactions on Machine Learning Research (**TMLR**)*, 2024 [CoLLAs-2025 Event Certified] Link
8. **M.Y. Harun***, J. Gallardo*, T.L. Hayes, R. Kemker, and C. Kanan. SIESTA: Efficient Online Continual Learning with Sleep. *In: Transactions on Machine Learning Research (**TMLR**)*, 2023 [CoLLAs-2024 Event Certified] Link
9. **M.Y. Harun**, J. Gallardo, T.L. Hayes, and C. Kanan. How Efficient Are Today's Continual Learning Algorithms? *In: CVPR Workshop on Continual Learning in Computer Vision (**CLVISION**)*, 2023 Link
10. **M.Y. Harun**, M.A. Rahman, J. Mellinger, W. Chang, T. Huang, B. Walker, K. Hori, and A. Ohta. Image Segmentation of Zona-Ablated Human Blastocysts. *In: IEEE International Conference on Nano/Molecular Medicine and Engineering (**NANOMED**)*, 2019 Link
11. **M.Y. Harun**, T. Huang, and A. Ohta. Inner Cell Mass and Trophectoderm Segmentation in Human Blastocyst Images using Deep Neural Network. *In: IEEE International Conference on Nano/Molecular Medicine and Engineering (**NANOMED**)*, 2019 Link
12. T.T. Huang, T. Kosasa, B. Walker, C. Arnett, C.T. Huang, C. Yin, **M.Y. Harun**, H.J. Ahn, and A. Ohta. Deep Learning Neural Network Analysis of Human Blastocyst Expansion from Time-lapse Image Files. *In: Reproductive BioMedicine Online (**RBMO**)*, 2021 Link
13. T. Huang, B. Walker, **M.Y. Harun**, M.A. Rahman, J. Mellinger, W. Chang, and A. Ohta. Automated Computer Analysis of Human Blastocyst Expansion from Embryoscope Time-Lapse Image Files. *In: American Society for Reproductive Medicine (**ASRM**)*, 2019 Link

Technical Skills

- **Core Expertise:** Generative-AI, LLMs, MLLMs/VLMs, foundation models, vision-language integration, continual learning, training/inference efficiency, representation geometry, domain generalization
- **Systems:** DeepSpeed, FSDP, TorchDist, multi-GPU scaling, efficient data pipelines
- **Programming Languages**: Python, C, C++, MATLAB
- **Libraries**: Numpy, Scipy, Scikit-learn, Pandas, OpenCV, Huggingface, Faiss, Weights & Biases
- **Deep Learning Framework**: PyTorch
- **OS & Applications**: Linux, MS Office, Git, Bash Scripting, LaTeX
- **Typing Speed**: 50 words per minute

Awards & Scholarships

- Best Poster Award — WNYISPW 2023
- Research Excellence Award: University of Hawaii at Manoa, 2020
- Teaching Excellence Award: University of Hawaii at Manoa, 2019

Invited Talks

- Controlling Neural Collapse Enhances Out-of-Distribution Detection and Transfer Learning, Oral Presentation at *ICCV 2025 Workshop on Building Foundation Models You Can Trust (TrustFM)*
- Towards Efficient Continual Learning in Deep Neural Networks, Invited Talk at *RIT Center for Human-aware Artificial Intelligence (CHAI) Seminar Series*, 2023

Teaching Experience

- Teaching Assistant: Undergraduate Course – Fourier Method for Imaging (RIT), Duration: Jan. 2021 – May 2021
- Teaching Assistant: Undergraduate Course – Imaging Science Fundamentals (RIT), Duration: Aug. 2020 – Dec. 2020
- Teaching Assistant: Undergraduate Course - C Programming Language (UHM), Duration: Aug. 2019 – Dec. 2019
- Mentor: Native Hawaiian Science and Engineering Mentorship Program, UHM, May 2019 – Aug. 2019
- Teaching Assistant: Undergraduate Course - Basic Circuit Lab (UHM), Duration: Jan. 2019 – May 2019
- Teaching Assistant: Undergraduate Course - Communication Systems Lab (UHM), Duration: Aug. 2018 – Dec. 2018

Reviewer

- Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- International Conference on Machine Learning (ICML), 2025
- ICML Workshop on Unifying Data Curation Frameworks Across Domains (DataWorld), 2025
- ICML Workshop on the Impact of Memorization on Trustworthy Foundation Models (MemFM), 2025
- IEEE ICDM Workshop on Open World Anomaly Detection in Dynamic and Evolving Environments (OWAD), 2025
- NeurIPS Workshop on Continual and Compatible Foundation Model Updates (CCFM), 2025
- Neural Information Processing Systems (NeurIPS), 2024
- Conference on Lifelong Learning Agents (CoLLAs), 2024-25
- European Conference on Computer Vision (ECCV), 2024
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2023-24
- CVPR Workshop on Continual Learning in Computer Vision (CLVISION), 2024
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2023
- IEEE International Conference on Robotics and Automation (ICRA), 2021

Service

- Research Trainee: AWARE-AI NSF Research Traineeship (NRT) Program, RIT, Jan. 2022 – May 2022
- Program Assistant: Vertically Integrated Projects (VIP), University of Hawaii at Manoa, 2020
- Organizer: Vertically Integrated Project (VIP) Exhibition, University of Hawaii at Manoa, 2018-2020
- Volunteer: IEEE International Conference on Nano/Molecular Medicine and Engineering (NANOMED), 2018
- Student Affiliate: East-West Center, University of Hawaii at Manoa, 2018-2020